

Chapter 13 - Binomial distribution

Counting successes in a fixed number of yes/no trials is one of the most common exam patterns in WU BBE mathematics. This chapter builds the binomial model from Bernoulli trials through the probability mass function, mean and variance, tails and complements, comparisons of two sides, and the approximations you meet on harder tasks - with worked examples at each step so the formulas become habits rather than slogans.

Learning objectives

- Recognise a Bernoulli trial and the four conditions of a binomial setting.

- Compute

13.1 Bernoulli trials

A Bernoulli trial is a single experiment with exactly two outcomes:

- success, with probability p

- failure, with

Typical labels: correct/incorrect, germinate/not, convert/not, defect/ok. The name "success" is only a label for the outcome you are counting - it need not be desirable. A "success" can be a defective item if that is what X is counting.

If Y is the indicator of success on one trial ($Y = 1$ on success, $Y = 0$ on failure), then

$$E[Y] = p, \quad \mathrm{Var}(Y) = p(1-p).$$

Variance $p(1-p)$ is largest when $p = 1/2$ and shrinks toward 0 as p approaches 0 or 1. That single fact explains many true/false traps about which of two people has the more variable score.

Example - one free throw

A player m

Example - variance peaks at $1/2$

Compare p

13.2 The binomial setting

Let X be the number of successes in n trials. We write

$$X \sim \mathrm{Bin}(n, p)$$

when all of the following hold:

1. Fixed n - the number of trials is set in advance.

2. Two outc

If any condition fails (for example, sampling without replacement from a tiny population so that p drifts, or two production lines with different defect rates forced into one "pooled" binomial), the plain $\mathrm{Bin}(n,p)$ model is the wrong story.

Exam habit

Before com
these.

Example - multiple-choice exam

A quiz has
question ha

Example - when it fails

Drawing 5
after each
A large-pop

13.3 Probability mass function

For $k = 0, 1, \dots, n$,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Two pieces multiply:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts how many sequences of length n have exactly k successes.

- $p^k(1-p)^{n-k}$

Special cases worth memorising:

Raising a probability below $1/2$ to a large power shrinks it extremely fast - perfect-run claims near a round percentage are often designed as close calls.

Example - three coin flips, exactly two heads

Fair coin: n

$$P(X = 2) = \binom{3}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1 = 3 \cdot \frac{1}{8} = \frac{3}{8}.$$

The three sequences are HHT, HTH, THH - each with probability $1/8$.

Example - guesser gets all 10 correct

$X \sim \mathrm{ma}$

$$P(X = 10) = (0.25)^{10} \approx 9.54 \times 10^{-7} \approx 0.000095\%,$$

far below 0.01% . Near a round cutoff, compute p^n exactly - do not eyeball.

Example - building $\binom{n}{k}$

$\binom{10}{k}$

13.4 Mean, variance, and standard deviation

Write $X = Y_1 + \dots + Y_n$ as a sum of independent Bernoulli indicators with the same p . Linearity and independence give

$$E[X] = np,$$

$$\mathrm{Var}(X) = np(1-p),$$

$$\mathrm{SD}(X) = \sqrt{np(1-p)}.$$

Useful comparisons when two sides share the same n :

- Mean difference: $E[B] - E[A] = n(p_B - p_A)$.

- Mean ratio

Trap

A higher sum variable than

Example - expected correct answers

Guesser: n

Example - who is more variable?

Same example

Example - mean vs a pass bar

Inspector v

13.5 Cumulative probabilities and English tails

Exam statements almost never stop at "exactly k ". Translate the English carefully:

The outcomes $X = k, k+1, \dots, n$ are mutually exclusive, so an "at least" probability is literally a sum of binomial terms - not one term and not p itself.

Complement shortcuts save time on long tails:

$$P(X \geq k) = 1 - P(X \leq k-1),$$

$$P(X \leq k) = 1 - P(X \geq k+1).$$

Example - "at least 7 out of 10"

Pass if $X \geq 7$

$$P(X \geq 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10) = \sum_{x=7}^{10} \binom{10}{x} p^x (1-p)^{10-x}.$$

A single term $P(X = 7)$ would undercount. On true/false items, check that the written sum starts at the threshold and ends at n .

Example - at least one success

$n = 5, p = 0.8$

$$P(X \geq 1) = 1 - (0.8)^5 = 1 - 0.32768 = 0.67232.$$

Computing $P(1) + \dots + P(5)$ works but is slower; the complement is the exam shortcut.

Example - "more than 8" vs "at least 8"

For $n = 10$
starting ind

13.6 Comparing two binomial sides

Many BBE-style tasks put two people, machines, or batches side by side with the same n but different p (or occasionally different n). Typical claim types:

1. Perfect run - check p^n against a percentage cutoff; compute precisely near round thresholds.

2. Tail wor

Ratio discipline

"B is at le
Compute b

Example - free throws, clutch bonus

$n = 10$ sho

Example - ratio trap

Suppose P

$$\frac{P_B}{P_A} \approx \frac{98.72}{0.35} \approx 282.$$

A claim "at least 300 times as likely" is false even though 282 is "near" 300. Do not round the ratio up to the exam's round number.

Example - same wording, different p

Statement:

13.7 When a sum is not binomial

If $X_1 \sim \text{Bin}(n_1, p_1)$ and $X_2 \sim \text{Bin}(n_2, p_2)$ are independent, then

$$X_1 + X_2 \sim \text{Bin}(n_1 + n_2, p)$$

only when $p_1 = p_2 = p$.

If $p_1 \neq p_2$, the total is a Poisson-binomial count (independent Bernoullis with unequal success probabilities). You may still use linearity:

$$E[X_1 + X_2] = n_1 p_1 + n_2 p_2,$$

$$\mathrm{Var}(X_1 + X_2) = n_1 p_1(1-p_1) + n_2 p_2(1-p_2).$$

A "pooled" rate $\bar{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ matches the mean of $\mathrm{Bin}(n_1+n_2, \bar{p})$, but the variances generally differ - equal means do not make the distributions equal.

Example - two production lines

Line A: n_1

Example - when pooling is fine

Two identical
the key.

13.8 Poisson approximation

When n is large, p is small, and $\lambda = np$ stays moderate,

$$P(X = k) \approx e^{-\lambda} \frac{\lambda^k}{k!}, \quad X \approx \mathrm{Poisson}(\lambda).$$

Use this for rare events (defects, mutations, server errors) when a task invokes the Poisson model. Check that p is small and n is large before trusting the approximation on an exam claim.

Example - rare defects

$n = 200$ iter

$$P(X = 0) \approx e^{-2} \approx 0.1353, \quad P(X = 1) \approx e^{-2} \cdot 2 \approx 0.2707.$$

Exact binomial values are close; the Poisson formula is much faster by hand.

Example - when not to use Poisson

$n = 10, p =$
not Poisson

13.9 Normal approximation

For large n with both np and $n(1-p)$ at least about 5,

$$X \approx \mathcal{N}(np, np(1-p)).$$

Because X is discrete and the normal is continuous, tail comparisons usually need a continuity correction, for example

$$P(X \geq k) \approx P\left(Y \geq k - \frac{1}{2}\right)$$

for $Y \sim \mathcal{N}(np, np(1-p))$. Always verify the np and $n(1-p)$ rule of thumb on each side before accepting a normal-approximation claim.

Example - rule of thumb check

$n = 40, p =$

Example - continuity correction sketch

$X \sim \text{ma}$

$$z = \frac{24.5 - 20}{\sqrt{12}} \approx 1.30.$$

Then use a standard normal table or calculator for $P(Z \geq 1.30)$.

13.10 Skewness (shape)

Binomial skewness is

$$\gamma_1 = \frac{1 - 2p}{\sqrt{np(1-p)}}.$$

- If $p < 1/2$, the distribution is skewed right (long right tail).

- If $p > 1/2$,

As n grows with p fixed away from the extremes, skewness shrinks and the shape becomes more bell-like - consistent with the normal approximation.

Example - shape from p

$n = 20, p =$
(skewed right)

13.11 Calculator and exam workflow

On TI-style calculators used in BBE practice:

- $\text{binompdf}(n, p, k) \rightarrow P(X = k)$

- binomcdf

Then build other tails from complements, for example $P(X \geq k) = 1 - \text{binomcdf}(n, p, k-1)$.

Example - calculator path for a pass probability

$n = 15, p =$

$$P(X \geq 13) = 1 - P(X \leq 12) = 1 - \text{binomcdf}(15, 0.6, 12).$$

Or sum binompdf at 13, 14, and 15 - same answer, more keystrokes.

A reliable attack order for true/false statements:

1. Confirm the binomial setting (n , p , independence, two outcomes).

2. Translate

Example - full mini-case

Agent A: n

13.12 Quick formula sheet

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E[X] = np, \quad \mathrm{Var}(X) = np(1-p), \quad \mathrm{SD}(X) = \sqrt{np(1-p)}$$

$$P(X \geq 1) = 1 - (1-p)^n, \quad P(X = n) = p^n, \quad P(X = 0) = (1-p)^n$$

$$\text{Poisson: } \lambda = np; \quad \text{Normal: } X \approx \mathcal{N}(np, np(1-p))$$

$$\gamma_1 = \frac{1-2p}{\sqrt{np(1-p)}}$$

When you can state the setting, write the PMF, move fluently between English and sums, work a short numerical example for means/variances/tails, and handle approximations without mixing the tools, you are ready for the practice tasks in this chapter.